Francisco Villacis

CONTACT INFORMATION

Department of Pure Mathematics
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PUBLICATIONS

Elana Kalashnikov, Ruxandra Moraru, Francisco Villacis, *Abelianization and moment maps* (preprint). arXiv:2610.42123
Francisco Villacis, *Principal minors and absolute values* (preprint). arXiv:2609.25499

EDUCATION

University of Waterloo

Ph.D. in Pure Mathematics, 2022 – 2026 (Expected).

University of Waterloo

M.Math. in Pure Mathematics. 2021 – 2022.

- CGPA: 92.50 (out of 100)

Carleton University

B.Sc. in Mathematics and Physics (Double Honours). 2017 – 2021.

- Overall CGPA: 11.30 (out of 12)
- Major CGPA: 11.52 (out of 12)

HONOURS AND AWARDS

Queen Elizabeth II Graduate Scholarship in Science & Technology (QEII-GSST). 2025 – 2026

TEACHING

University of Waterloo

MATH 127 – Calculus 1 for the Sciences, Spring 2025

TALKS GIVEN

University of Waterloo

Linear amoebas and their combinatorial properties
Algebraic Geometry Working Seminar 2026

Matroids, convex polyhedra, and stratifications of the Grassmannian
Algebraic Geometry Working Seminar 2025

Computing the quantum cohomology of projective space and of the Grassmannian
Mirror Symmetry Seminar 2025

D-modules on smooth varieties
Chiral de Rham Seminar 2023

RESEARCH INTERESTS

I study the images and fibres of restricted toric moment maps in symplectic and algebraic geometry. My doctoral work uses matroid theory to describe compactified linear amoebas by finitely many circuit points. I am interested in which configurations of these points arise from complex linear spaces, how the points 'collide' under degeneration, and what information they retain in the tropical limit.

EXPERIENCE

University of Waterloo

Fall	2026	Teaching Assistant, Calculus 1 for Honours Math. (MATH 137).
Fall	2025	Teaching Assistant, Calculus 1 for Honours Math. (MATH 137).
Fall	2025	Teaching Assistant, Multivariable Calculus - Advanced Level (MATH 247)
Summer	2025	Instructor, Calculus 1 for the Sciences (MATH 127).
Winter	2025	Teaching Assistant, Modern Algebraic Geometry (PMATH 945).
Winter	2025	Teaching Assistant, Applied Complex Analysis (PMATH 332).
Winter	2025	Teaching Assistant, Calculus 2 for Honours Math. (MATH 138).
Fall	2024	Teaching Assistant, Commutative Algebra (PMATH 646).
Fall	2024	Teaching Assistant, Linear Algebra for Engineers (MATH 115).
Spring	2024	Teaching Assistant, Proofs in Geometry (MATH 661).
Spring	2024	Teaching Assistant, Real Analysis (PMATH 351).
Winter	2024	Teaching Assistant, Complex Analysis (PMATH 352).
Winter	2024	Teaching Assistant, Calculus 2 for the Sciences.
Fall	2023	Teaching Assistant, Smooth Manifolds (PMATH 465/665).
Fall	2023	Tutor for Multivariable Calculus and Advanced Linear Algebra.
Summer	2023	Tutor for Multivariable Calculus and Advanced Linear Algebra.
Winter	2023	Teaching Assistant, Algebraic Topology (PMATH 667).
Winter	2023	Teaching Assistant, Calculus 2 - Advanced Level (MATH 148)
Fall	2022	Teaching Assistant, Calculus 3 for Honours Math (MATH 237).
Fall	2022	Tutor for Multivariable Calculus and Advanced Linear Algebra.
Spring	2022	Teaching Assistant, Calculus 2 for Honours Math (MATH 138).
Spring	2022	Teaching Assistant, Intro to Group Theory (PMATH 336).
Winter	2022	Teaching Assistant, Differential Geometry (PMATH 365).
Winter	2022	Teaching Assistant, Linear Algebra for Honours Mathematics.
Fall	2021	Teaching Assistant, Calculus 1 for Engineering.
Fall	2021	Teaching Assistant, Calculus 1 for the Sciences.

Carleton University

Fall	2020	Teaching Assistant, Algebra.
Fall	2020	Teaching Assistant, Elementary Calculus I.
Summer	2020	Students as Partners Program, Algebra.
Summer	2020	Teaching Assistant, Multivariable Calculus for Engineering or Physics.
Summer	2020	Teaching Assistant, Intermediate Calculus.
Winter	2020	Teaching Assistant, Multivariable Calculus for Engineering or Physics.
Winter	2020	Teaching Assistant, Linear Algebra for Engineering or Science.
Fall	2019	Teaching Assistant, Calculus for Engineering or Physics.
Fall	2019	Teaching Assistant, Intermediate Calculus.
Fall	2019	Teaching Assistant, Algebra.
Summer	2019	Research Assistant, Hyper-Kamiokande Collaboration.

Relevant Skills

- Native Spanish speaker and fluent English and French speaker.
- Python (NumPy, pandas, SciPy, PyTorch), Mathematica, MATLAB.
- Simulation, Monte Carlo techniques.